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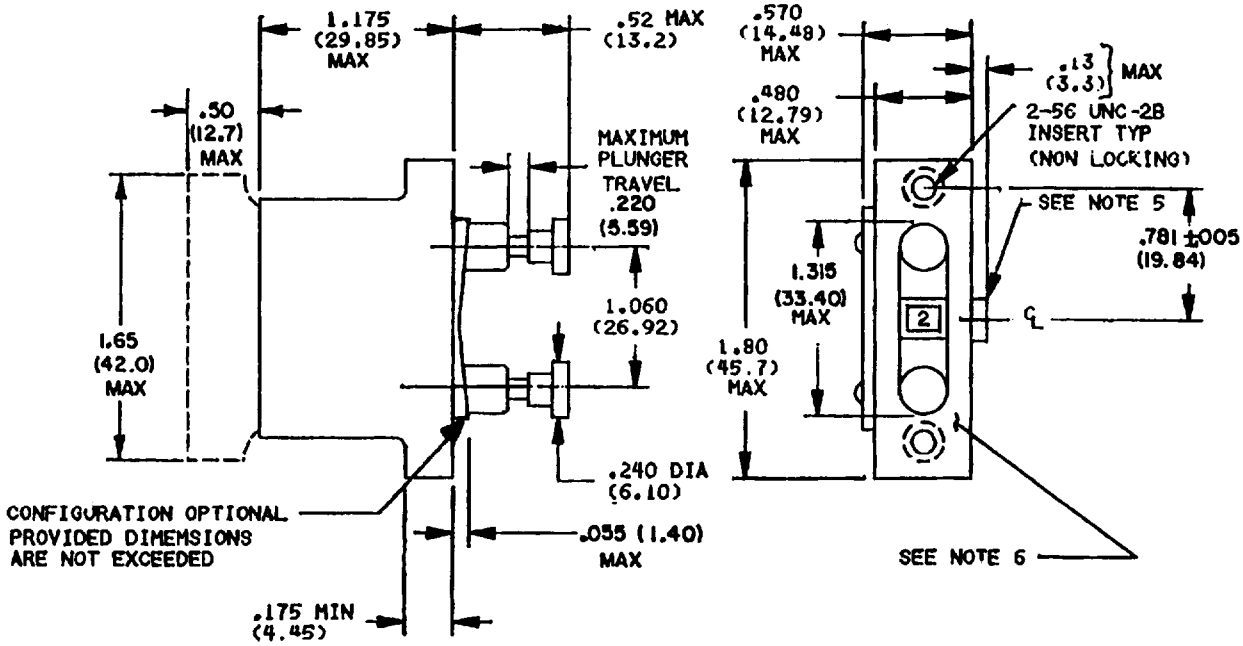
MIL-PRF-22710/30
25 November 1983

PERFORMANCE SPECIFICATION SHEET

SWITCHES, ROTARY (PRINTED CIRCUIT), PUSHBUTTON BIDIRECTIONAL, SEALED, HIGH SHOCK, RFI SHIELDED, ILLUMINATED OR NONILLUMINATED, LOW LEVEL, STYLE SRPC30

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-22710.

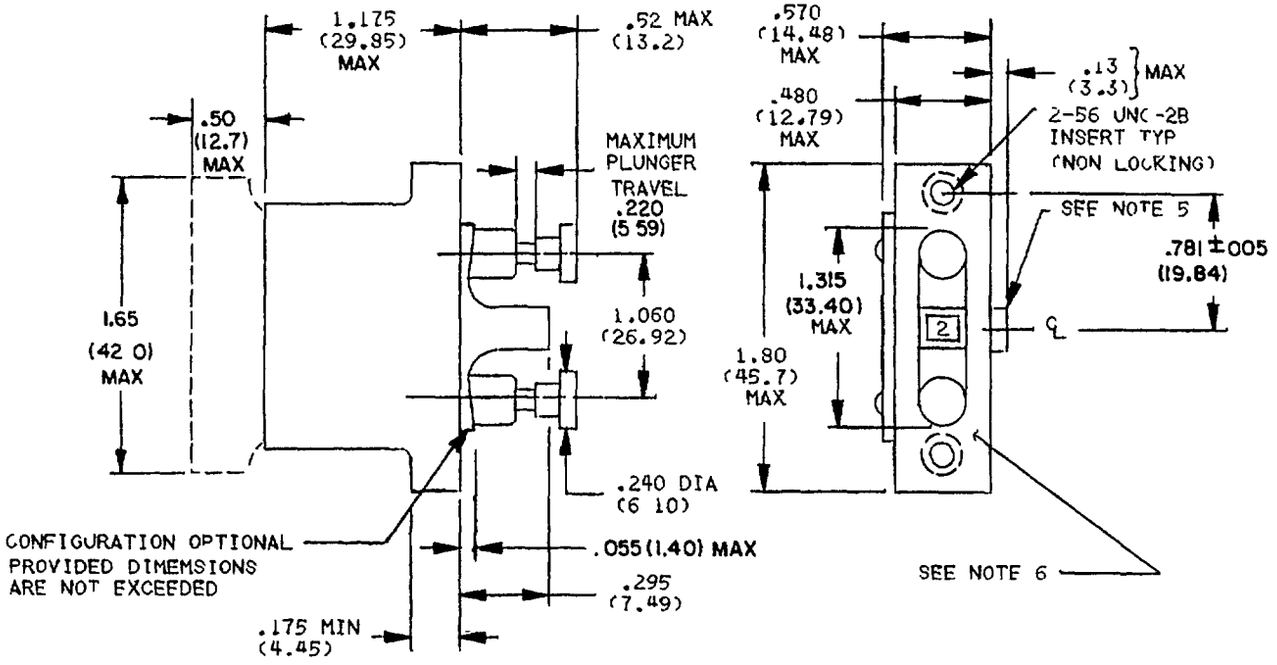


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
4. Unless otherwise specified, tolerance are ±.03 (.8 mm) for two place decimals and ±.010 (.25 mm) for three place decimals.
5. Lamps installed here, where illumination is specified.
6. This mounting surface has optional conductive coating to provide RFI shielding, when required.
7. Depressing top botton increases readout digit. Depressing lower button decreases readout digit.

FIGURE 1. Flush lens style.

FSC 5930

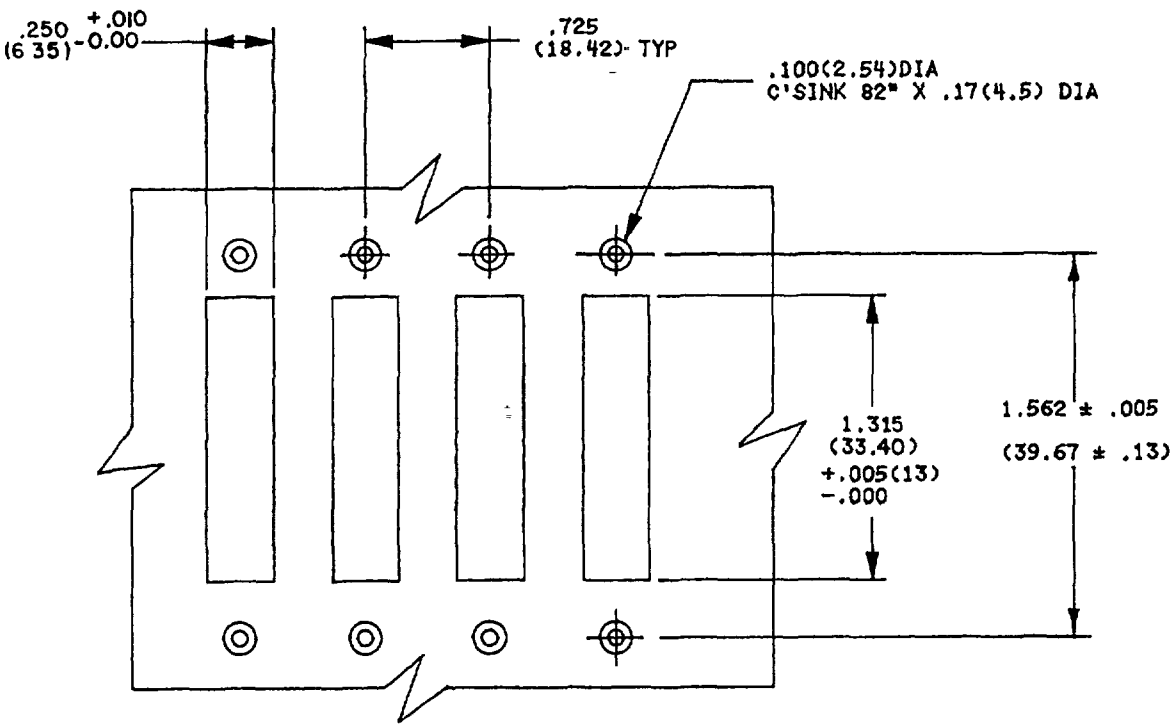
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4. Unless otherwise specified, tolerance are ±.03 (.8 mm) for two place decimals and ±.010 (.25 mm) for three place decimals.
5. Lamps installed here, where illumination is specified.
6. This mounting surface has optional conductive coating to provide RFI shielding, when required.
7. Depressing top botton increases readout digit. Depressing lower button decreases readout digit.

FIGURE 2. Extended lens style.

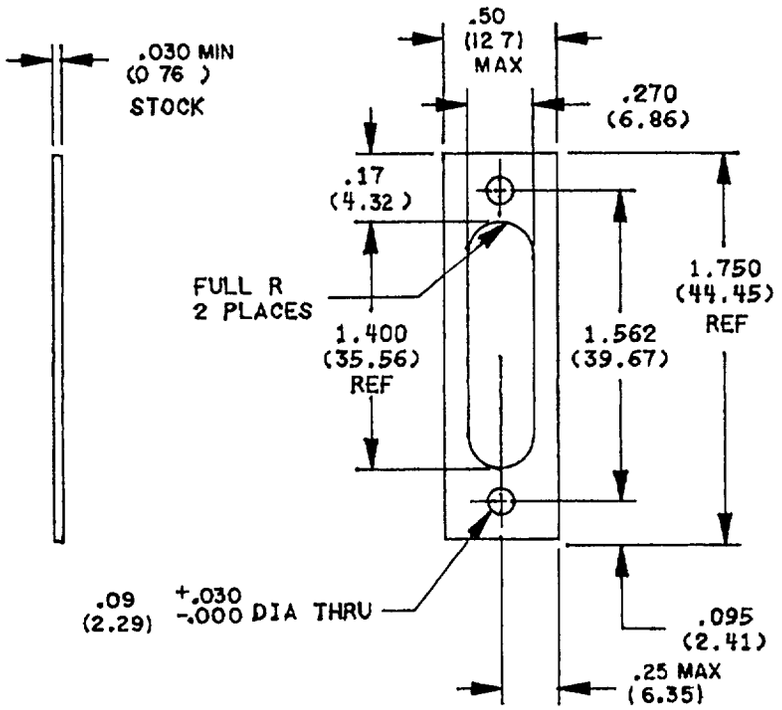
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NOTES:
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2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
4. Unless otherwise specified, tolerance are ±.03 (.8 mm) for two place decimals and ±.010 (.25 mm) for three place decimals.
5. Panel cutout is applicable to both flush lens and extended lens styles.

FIGURE 4. Recommended panel cutout.

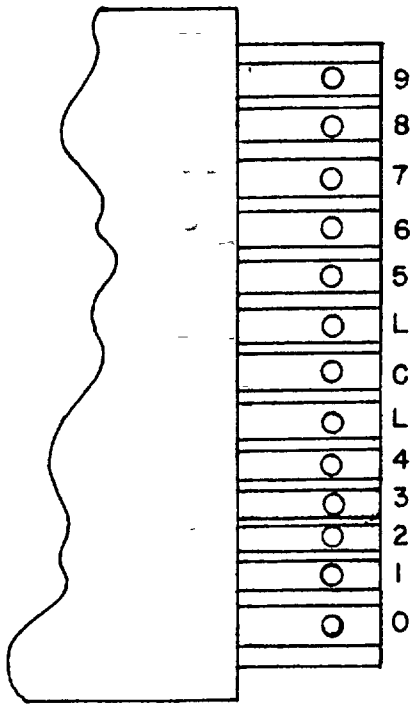
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4. Unless otherwise specified, tolerances are ±.03 (.8 mm) for two place decimals and ±.010 (.25 mm) for three place decimals.

FIGURE 3. Panel seal gasket.

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.032 (.81) minimum Thk PCB; .038 (.97) minimum dia. solder hole connections.

Module A

Truth table I

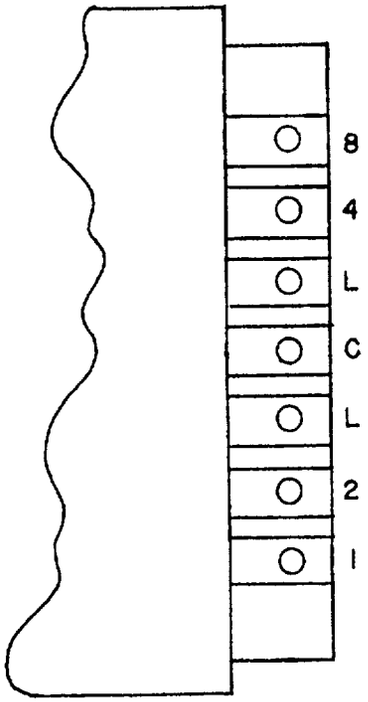
(Circuit configuration 1)

36° Indexing

DIAL READS	Common C Connected to:
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9

FIGURE 5. Terminal design and circuit configuration.

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.032 (.81) minimum Thk PCB; .064 (1.63) minimum dia. solder hole connections.

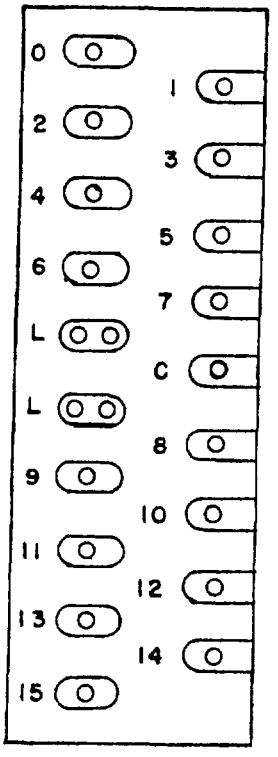
Module B

Truth table 2
(Circuit configuration 2)
36° Indexing

DIAL LEADS	Common C Connected to:			
	1	2	4	8
0				
1	●			
2		●		
3	●	●		
4			●	
5	●		●	
6		●	●	
7	●	●	●	
8				●
9	●			●

FIGURE 5. Terminal design and circuit configuration - Continued.

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Truth table 3
(Circuit configuration 3)
22.5° Indexing

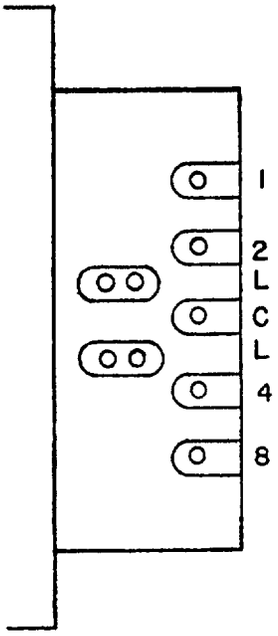
DIAL READS	Common C Connected to:
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14
15	15

.032 (.81) minimum Thk PCB; .038 (.97) minimum dia. solder hole connections.

Module C

FIGURE 5. Terminal design and circuit configuration - Continued.

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.032 (.81) minimum Thk PCB; .038 (.97) minimum dia. solder hole connections.

Module D

Truth table 4

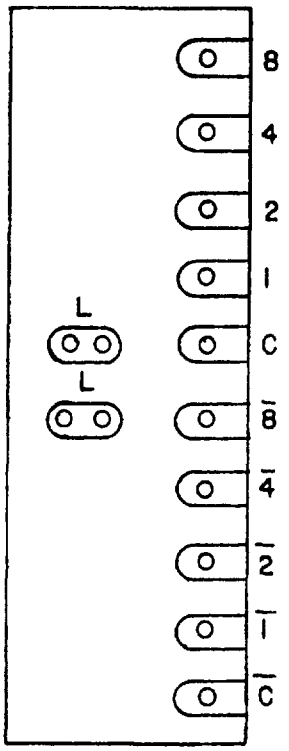
(Circuit configuration 4)

22.5° Indexing

DIAL READS	Common C Connected to :			
	1	2	4	8
0				
1	●			
2		●		
3	●	●		
4			●	
5	●		●	
6		●	●	
7	●	●	●	
8				●
9	●			●
10		●		●
11	●	●		●
12			●	●
13	●		●	●
14		●	●	●
15	●	●	●	●

FIGURE 5. Terminal design and circuit configuration - Continued.

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.032 (.81) minimum Thk PCB; .038 (.97) minimum dia. solder hole connections.

Module E

Truth table 5
(Circuit Configuration 5)
22.5° Indexing

DIAL READS	Common C Connected to :				Common $\overline{C}$ Connected to :			
	1	2	4	8	$\overline{1}$	$\overline{2}$	$\overline{4}$	$\overline{8}$
0					●	●	●	●
1	●					●	●	●
2		●			●		●	●
3	●	●					●	●
4			●		●	●		●
5	●		●			●		●
6		●	●		●			●
7	●	●	●					●
8				●	●	●	●	
9	●			●		●	●	
10		●		●	●		●	
11	●	●		●			●	
12			●	●	●	●		
13	●		●	●		●		
14		●	●	●	●			
15	●	●	●	●				

NOTES :

1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only and are based on 1 inch equal to 25.4mm.

FIGURE 5. Terminal design and circuit configuration - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figures 1, 2, and 5.

Construction: D.

Indexing: 22.5 or 36 degrees.

Temperature-life characteristic: D, test loads as shown in table I (For illuminated switches, maximum temperature is 85°C).

TABLE I. Electrical test load.

Test or condition	Resistive	
At atmospheric pressure	0.125 A	28 V dc and 115 V ac
At reduced barometric pressure	0.125 A	28 V dc and 115 V ac
Life, low level	10 mA	30 mV

Vibration grade: 3 (15g, 10-2,000 Hz).

Shock type: M(100g) and H(high impact). Methods I and II are applicable.

Altitude rating: C (up to 70,000 feet).

Terminal design: See applicable detail on figure 5 and table III.

Contact resistance:

Initial - 100 milliohms, maximum.
Following life (rotational) and moisture resistance - 60 milliohms, maximum above initial value.

Operating force (push force required to index from any detent position to the next):

20 ounces, minimum.
40 ounces, maximum.

Seal: Method II.

Dielectric withstanding voltage:

At atmospheric pressure: 750 volts, rms for one minute.
At reduced barametric pressure: 250 volts, rms.

Sand and dust: Applicable.

Explosion: Applicable.

Life (rotational): MIL-STD-202, method 206, test condition G.

Contact rating: .125 ampere at 28 volts (ac peak to peak, or dc) Resistive.

Moisture resistance: Method II.

Luminance: Illuminated switches are as shown in table II. Lamps are supplied with the switches.

Marking: Terminals shall be marked as shown on figure 5.

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TABLE II. Luminance.

Lamp	Switch filter lens	Brightness	Rating	
			Volts	Amperes
MS90542-6833	Clear	2.0 ±1.0 Foot-lamberts	5	.115
MS90542-6833	Red	1.0 ±.05 Foot-lamberts	5	.115

RFI shielded: When specified, see table III.

Life, low-level switching: Applicable.

Qualification: Qualification of the RFI shielded version of a particular switch type shall qualify the non-RFI shielded unit.

Part number: M22710/30- X X X

Style and RFI option: (see table III)

Truth tables I thru 5: (see figure 5.)

Luminance option: (see table IV)

TABLE III. Style and RFI option.

Code	Style	RFI shielded
A	Flush lens	No
B	"	Yes
C	Extended lens	No
D	"	Yes

TABLE IV. Luminance option.

Code	Illum.	Marking	Background
A	None	White opaque	Black opaque
B	Clear	White translucent	"
C	Red	"	"
D	None	Black opaque	White opaque
E	Clear	"	White translucent
F	Red	"	"

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Navy - EC
Air Force - 85

Review activities:
Army - SM
Air Force - 99
DLA - ES

User activities:
Army - ME, AL
Navy - MC, AS
Air Force - 11, 17

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5930-1192)

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a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

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